

LAB # 05b

Implementation of switch case statement in java

Objective

Implementation of switch case statement in java.

Lab Goals

To provide working insight into the usage of switch case statement in java programming:

- a. Switch Statement.

Introduction to switch case statement

In Java, the switch statement is a control flow statement that allows you to select one of many code blocks to execute based on the value of a variable or an expression. It provides a concise way to handle multiple cases and avoid long chains of if-else statements. It has the following syntax.

Syntax

```
switch (expression)
{
    case value1:
        statement1;
        break;
    case value2:
        statement2;
        break;
    case valueN:
        statementN;
        break;
    default:
        statementDefault;
}
else
{
    // Executes this block if
    // condition is false
}
```

Here's the syntax of a switch statement:

- In the switch statement, the expression is evaluated, and its value is compared against each case value.
- If a case value matches the value of the expression, the corresponding code block is executed. The break statement is used to terminate the switch block and exit the statement.
- If a case doesn't have a **break statement**, the execution continues to the next case until a break is encountered or the switch block ends.
- The default case is optional and represents the code block to be executed when none of the case values match the expression. It acts as a fallback option when no other cases match.

Example#1

```
// Java program to illustrate switch-case
class SwitchCaseDemo {
    public static void main(String args[]) {
        int i = 9;
        switch (i) {
            case 0:
                System.out.println("i is zero.");
                break;
            case 1:
                System.out.println("i is one.");
                break;
            case 2:
                System.out.println("i is two.");
                break;
            default:
                System.out.println("i is greater than 2.");
        }
    }
}
```

Output

i is greater than 2.

Example#2

```
public class Main {  
    public static void main(String[] args) {  
        String fruit = "apple";  
        int price;  
        switch (fruit) {  
            case "apple":  
                price = 10;  
                break;  
            case "banana":  
                price = 15;  
                break;  
            case "orange":  
                price = 12;  
                break;  
            default:  
                price = 0;  
                break;        }  
        System.out.println("Price of " + fruit + " is " + price);    }  
}
```

Output

```
Price of apple is 10
```

Example#3

```
public class Main {  
    public static void main(String[] args) {  
        long number = 1000L;  
        String numType;  
        switch ((int)(number % 2)) {  
            case 0:  
                numType = "Even";  
                break;  
            case 1:  
                numType = "Odd";  
                break;  
            default:  
                numType = "Invalid";  
                break;  
        }  
        System.out.println(number + " is " + numType);  
    } } }
```

Output

```
1000 is Even
```

Exercise

A. Write a program using switch which choose between following cases:

- a: Add two numbers
- b: Find power using math function
- c: Exit

Expected Output:

```
Choose an option:
1. Say Hello
2. Say Goodbye
3. Exit
1
Hello!
```

B. Write a program in Java to calculate the total cost of items purchased based on their category. The program should prompt the user to enter the category of the item (1 for electronics, 2 for groceries, and 3 for clothing) and the amount spent on that category. Then, calculate and display the total amount spent on each category separately. Use a switch-case statement to implement this program.

Expected Output:

```
Enter the category of item:
1. Electronics
2. Groceries
3. Clothing
2
Enter amount spent: 75.50
Total spent on Groceries: $75.5
```

- C. Write a program in Java to convert a numerical grade into a letter grade. The program should prompt the user to enter a numerical grade between 0 and 100 and then display the corresponding letter grade based on the following grading scale:

- 90-100: A
- 80-89: B
- 70-79: C
- 60-69: D
- Below 60: F

Use a switch-case statement to implement this program.

Hint: grade should be user input for suppose 88 you have taken as user input.

Then divide it **int range = grade / 10;** after that use **range** as an expression in switch Case.

Expected Output:

```
Enter your grade (0-100): 88
Letter Grade: B
```

- D. Write a program in Java to calculate the area of different geometric shapes. The program should prompt the user to choose the shape (1 for circle, 2 for rectangle, and 3 for triangle) and then input the necessary parameters (radius for circle, length and width for rectangle, and base and height for triangle). Finally, calculate and display the area of the chosen shape. Use a switch-case statement to implement this program.

Formula:

1. **Area of Circle:** $\text{Math.PI} * \text{radius} * \text{radius}$
2. **Area of Rectangle:** $\text{length} * \text{width}$
3. **Area of Triangle:** $0.5 * \text{base} * \text{height}$

Expected Output:

```
Choose shape to calculate area:
1. Circle
2. Rectangle
3. Triangle
3
Enter base: 10
Enter height: 5
Area of Triangle: 25.0
```